

Football Match Outcome Prediction

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Abstract--- This study presents a time-aware machine learning pipeline to predict home team victories in the English Premier League. Given the highly stochastic nature of football, the problem was framed as a binary classification task. Six different algorithms (*k*-NN, Naive Bayes, SVM, Random Forest, XGBoost, and Artificial Neural Networks) were trained and evaluated using rolling window features derived from historical match statistics. The best predictive performance was achieved by the SVM model following Platt calibration (AUC: 0.6864, F1-Score: 0.5424). Additionally, the model's outputs were examined from an ethical and fairness perspective to quantitatively assess any systematic bias regarding the home advantage.

I. INTRODUCTION

A. Motivation and Problem Statement

Football is an incredibly dynamic and low-scoring sport where random events often dictate the final score. Because of this randomness, predicting match results remains a tough but interesting challenge in statistical machine learning. In this study, the main goal is to use historical pre-match statistics to predict if the home team will win in English Premier League matches.

Instead of the traditional three-class setup (Home/Draw/Away), the problem is framed as a binary classification task (Home Win vs. Not Win). Since predicting draws carries a high level of irreducible uncertainty and football has an asymmetric decision value, using a binary approach helps reduce noise and improves the model's overall reliability.

B. Paper Organization

The rest of this paper is organized as follows: Section II introduces the dataset and covers both the exploratory data analysis (EDA) and feature engineering steps. Section III details the cross-validation strategy along with the classical and ensemble machine learning models used. Section IV presents the performance comparison of these models and their probability calibrations. Section V explores ethical considerations, specifically looking into model interpretability and home advantage bias. Finally, Section VI summarizes the main findings and suggests ideas for future work.

II. DATA & PREPROCESSING

A. Dataset Provenance

The dataset used in this study comes from Kaggle (sourced from Football-Data.co.uk) and includes English Premier League matches from the 2000/01 to 2024/25 seasons. It contains no personally identifiable information (PII) or protected demographic details, making it perfectly suitable for academic research. After the initial cleaning and a rolling window warmup period, the original 9,380 raw observations were reduced to 9,125 usable matches for modeling. These final samples are represented by 26 pre-match rolling features. To ensure reproducibility and prevent temporal leakage, the train-test split was performed strictly in chronological order. The training set covers the seasons between 2000 and 2020, while the test set spans from 2020 to 2025. The historical match data from Football-Data.co.uk is hosted on Kaggle under a CC0 Public Domain license. The final processed dataset contains 26 features, satisfying the course criteria by blending continuous variables (such as rolling shot differences and win-rate shifts) with discrete/categorical factors (such as venue indicators and categorized form tiers).

B. Exploratory Data Analysis

Looking at the target variable's distribution, there is a clear structural advantage for home wins along with a moderate class imbalance.

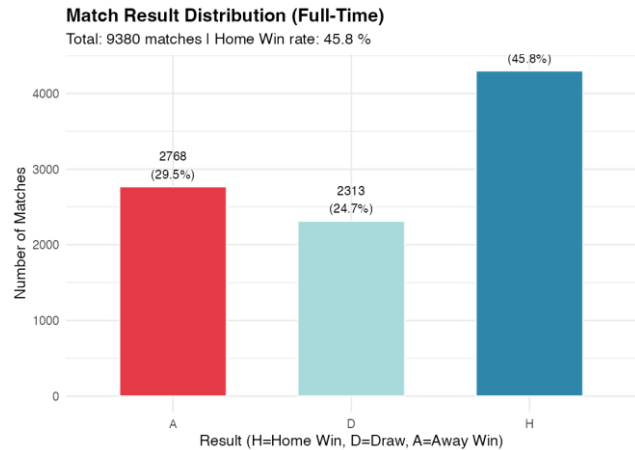


Fig. 1. Distribution of full-time match results highlighting the home win class imbalance

Seasonal trend analysis also shows that the home advantage fluctuates over time. For instance, there is a noticeable drop during unusual periods like the COVID-19 pandemic when games were played without spectators. Because of these temporal shifts, maintaining the strict chronological order of the dataset is highly crucial for the modeling phase.

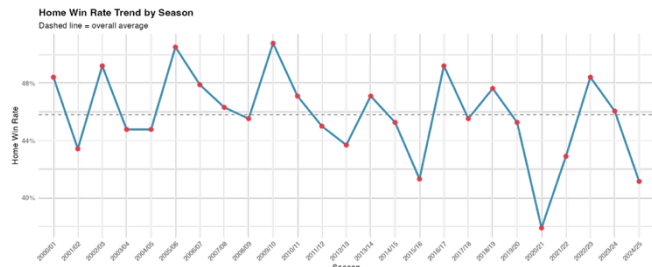


Fig. 2. Longitudinal trend of home win rates across Premier League seasons.

C. Feature Engineering and Preprocessing

To ensure the model is actually viable for real-world scenarios and live predictions, preventing data leakage was a top priority. Including in-game statistics (such as the number of shots on target or goals scored during that specific match) would cause severe leakage. Thus, a time-aware feature engineering approach was designed using only pre-match data.

To capture each team's current form, 5-match rolling window averages were calculated for metrics like shots, corners, and cards. A shift(1) operation was applied to exclude the actual match being predicted from these historical averages. Additionally, since teams generally perform differently at home versus away, the feature calculations are venue-aware. This means the home team's statistics are based solely on their past home games, and the away team's statistics come strictly from their past away games. Finally, z-score scaling was fitted entirely on the training set and then applied to the test set, completely eliminating any remaining risk of data leakage.

III. METHODOLOGY

A. Cross-Validation Strategy

Football data naturally follows a chronological order, meaning there is a strict time dependence between matches. For this reason, standard random k-fold cross-validation was replaced with an expanding window TimeSeriesSplit approach. This prevents temporal data leakage and better simulates how the model would be deployed in real life. In this setup, the training dataset grows in each fold, and the model is tested on the next chronological slice. Our sensitivity analysis proved that random splitting causes temporal leakage and leads to misleadingly optimistic results, whereas time-series splitting yields much more reliable generalization scores. Additionally, to handle the slight class imbalance in the dataset, class weights (cost-sensitive learning) were applied to all models during the training phase.

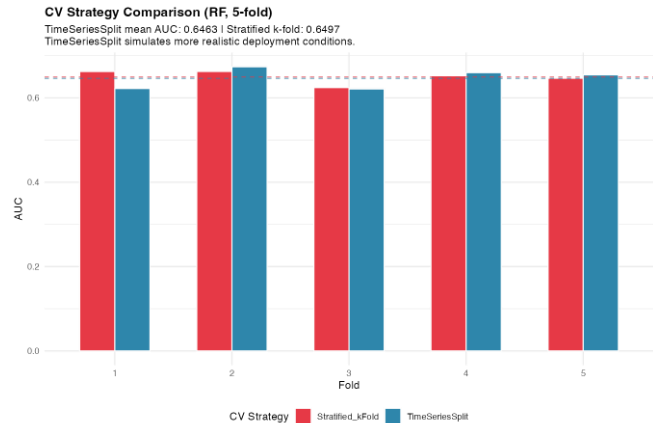


Fig. 3. Comparison of TimeSeriesSplit vs. Stratified k-fold performance to diagnose temporal leakage.

B. Classical Baselines

To establish a solid baseline for performance and analyze inductive biases, k-NN, SVM, and Naive Bayes models were selected. For the k-NN model, the neighborhood parameter (k) was optimized to 31 through time-series CV. Since k-NN relies entirely on geometric distances in the feature space without making parametric assumptions, it is highly sensitive to the curse of dimensionality and feature scales. Thus, the z-score normalization applied during preprocessing was a critical step for model stability. For the SVM, both linear and RBF kernels were tested. Since the underlying patterns in the data proved to be non-linear, the RBF kernel ($C=0.1$, $\gamma=0.001$) was chosen for providing the most stable decision boundary. The inductive bias of the SVM model relies on maximizing the geometric margin between classes while minimizing structural risk. It assumes that the noisy match data becomes optimally separable once projected into a higher-dimensional feature space via the RBF kernel. Finally, a Naive Bayes classifier was included. Although it strictly assumes conditional independence between features, it was tested to see how robust it would be against highly correlated variables like shot and corner counts, which clearly violate this assumption.

C. Ensemble Methods

Random Forest and XGBoost were implemented to compare how well Bagging and Boosting techniques perform on this dataset. The Random Forest model was configured with 500 trees and 4 features per split ($mtry=4$) to mainly reduce variance. On the other hand, XGBoost was aimed at minimizing bias through gradient boosting. After tuning, XGBoost's learning rate (η) was set to 0.1 and max depth to 3. To prevent overfitting, early stopping was utilized, which locked the optimal number of boosting rounds at 31. The empirical results demonstrated that while Random Forest's error naturally stabilized as more trees were added, XGBoost's low-bias advantage came with a severe overfitting risk when early stopping was ignored.

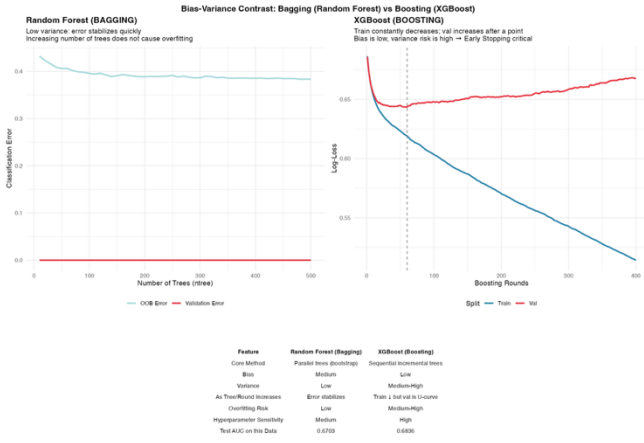


Fig. 4. Bias-Variance contrast analysis between Bagging (RF) and Boosting (XGBoost) methods.

D. Feedforward Neural Network

To capture complex and non-linear relationships in the dataset, a Feedforward Neural Network with a single hidden layer was constructed. Following cross-validation and hyperparameter tuning, the optimal number of hidden neurons (size) was set to 32. The single hidden layer uses 32 neurons with a Hyperbolic Tangent (tanh) activation function to handle the zero-centered continuous rolling differences, while the output layer applies a Sigmoid function to generate final probabilities. Weights were initialized using the Xavier method to keep variance stable during forward and backward propagation. Because football match data is inherently noisy, two main regularization strategies were applied to keep the model from simply memorizing the training set: an L2 weight penalty (weight decay = 0.001) and iterative early stopping. By analyzing the learning curves, it was observed that these regularization efforts successfully minimized the gap between training and validation losses, keeping the model's generalization capacity intact.

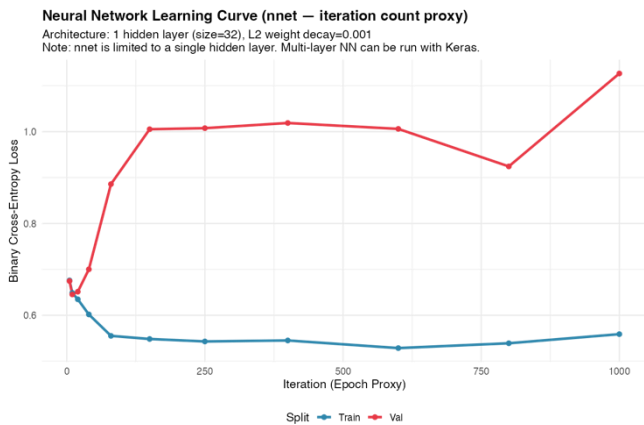


Fig. 5. Training vs. validation loss curve for the Feedforward Neural Network.

IV. EXPERIMENTS & RESULTS

A. Model Comparison and Evaluation

The trained models were evaluated on the chronologically split 20% test set using the default 0.50 decision threshold. Since Accuracy can be quite misleading when dealing with imbalanced data, the models were judged primarily on their F1-Score (for Precision-Recall balance) and ROC-AUC (for threshold-independent ranking). As shown in Table I, the Platt-calibrated Support Vector Machine (SVM) achieved the highest discriminative power, recording an AUC of 0.6864 and an F1-Score of 0.5424. The RBF kernel allowed this model to successfully capture the non-linear boundaries in the match data.

model_comparison_updated

Model	Calibration	AUC	Accuracy	Precision	Recall	F1_Score
SVM	Platt	0.6864	0.6395	0.6094	0.4887	0.5424
NaiveBayes	Raw	0.6849	0.6356	0.6118	0.4561	0.5226
XGBoost	Platt	0.6836	0.6356	0.6078	0.4699	0.53
RandomForest	Raw	0.6703	0.6367	0.6239	0.4261	0.5063
kNN	Raw	0.6541	0.6329	0.6151	0.4286	0.5052
NeuralNet	Platt	0.5748	0.5721	0.5114	0.4787	0.4945

Table 1. Performance Metrics of Predictive Model on the Test Set

A particularly interesting takeaway from these results is the performance of the Naive Bayes model. It secured second place with an AUC of 0.6849, outperforming heavy ensemble models like Random Forest (0.6703) and XGBoost (0.6836), as well as the distance-based k-NN (0.6541). Technically, highly correlated features like shots and corners violate the core conditional independence assumption of Naive Bayes. However, its strong performance empirically proves that in highly stochastic and noisy environments like football, complex tree-based algorithms are extremely prone to overfitting. On the contrary, simple probabilistic models with low variance tend to generalize much better. Unsurprisingly, the Single Layer Neural Network (NeuralNet) struggled the most (AUC: 0.5748), which is expected given the tabular nature of the data and the limited sample size.

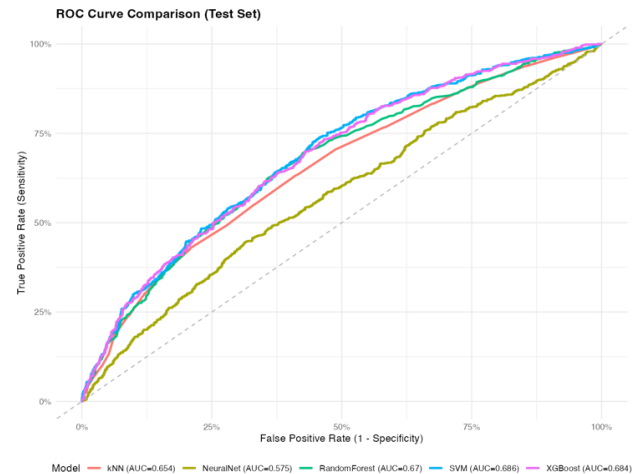


Fig. 6. Receiver Operating Characteristic (ROC) curves comparing the discriminative capacity of the calibrated models.

B. Probability Calibration and Uncertainty

To verify if the predicted probabilities actually align with real-world frequencies, Brier scores were calculated and reliability diagrams were plotted. The raw predictions were calibrated using both Isotonic Regression and Platt (Logistic) Scaling. For the SVM model, Platt scaling slightly reduced the Brier score from 0.2211 to 0.2208. This minor drop indicates that the SVM's raw probabilities were already quite well-calibrated.

To quantify the statistical uncertainty of the predictions, a Bootstrap approach with 500 iterations was applied. The resulting 95% Confidence Interval (CI) for the SVM's AUC was [0.6627, 0.7104], with a standard error of 0.0120. The model showed the highest uncertainty in matches where the predicted probabilities fell between 0.40 and 0.60. This makes perfect sense, as the underlying possibility of a draw heavily blurs the decision boundaries in football.

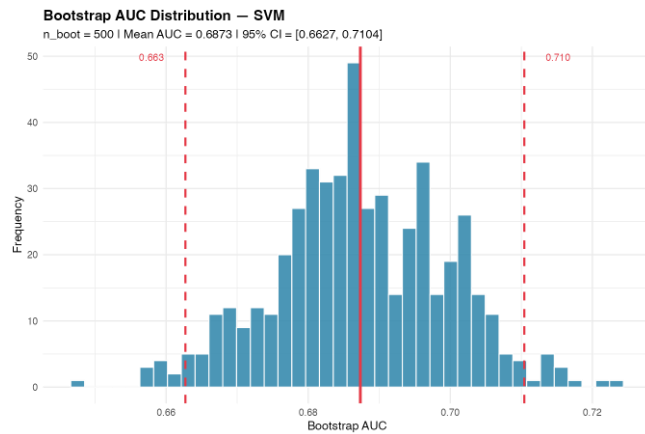


Fig. 7. Bootstrap distribution of the AUC score for the best performing SVM model (n=500 iterations).

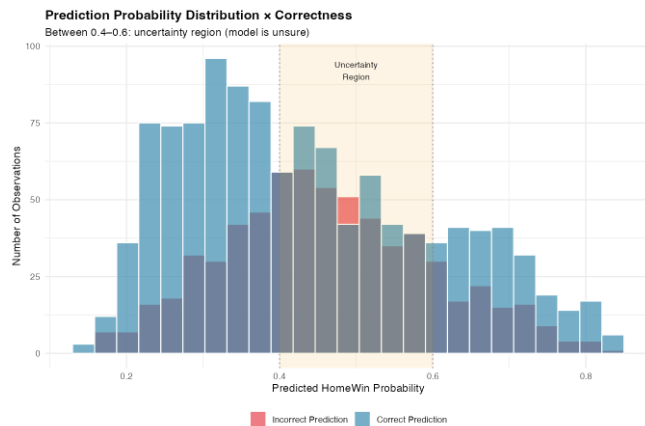


Fig. 8. Prediction probability distribution highlighted by correctness, revealing the 0.4-0.6 uncertainty region.

C. Threshold Analysis

While 0.50 is the standard default threshold for binary classification, class imbalances and unequal decision costs often make it suboptimal. To address this, various thresholds were tested to find the one that maximizes the F1-Score. The

analysis showed that the optimal decision threshold for the best-performing SVM model was actually 0.35 (F1: 0.6413). This lower threshold implies that the model is structurally conservative and hesitant to assign very high probabilities to the HomeWin class. Dropping the threshold from 0.50 to 0.35 did cause a slight bump in the False Positive rate, but it significantly improved the model's operational value by maximizing the number of actual home wins (Recall) correctly identified.

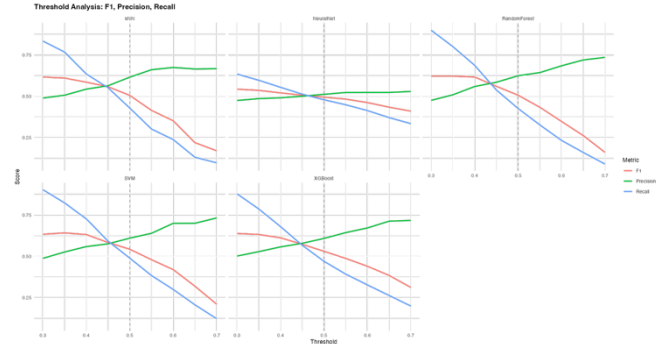


Fig. 9. Threshold tuning analysis demonstrating the maximization of the F1-Score at lower probability thresholds.

V. ETHICS: BIAS, FAIRNESS & INTERPRETABILITY

A. Interpretability with SHAP

High-performing ensemble models like XGBoost and Random Forest often act as black boxes. To ensure transparency in the decision-making process and to verify that the algorithm actually learned meaningful football dynamics, a SHAP (SHapley Additive exPlanations) analysis was applied. Even though the SVM achieved the highest overall performance, the SHAP analysis was conducted on XGBoost, which was the best-performing tree-based model. This decision was made because SHAP is natively defined for gradient boosted trees, offering a much clearer and more interpretable feature decomposition.

Looking at the SHAP values and Gain metrics, the three most critical features driving the predictions were the Shot Difference (roll_shot_diff), Win Rate Difference (roll_winrate_diff), and Shots on Target Difference (roll_sot_diff) between the home and away teams. Interestingly, Goal Difference (roll_goal_diff) ranked lower in importance compared to these shot-based metrics. This proves that the model is not merely memorizing scoreboard results, but rather learning the underlying tactical dominance and actual on-pitch pressure created by the teams.

While football analytics does not involve life-or-death risks like healthcare or criminal justice, making it a lower-stakes domain, it still holds financial weight in betting markets and club governance. This application presents a clear interpretability-accuracy trade-off. Our XGBoost model provides straightforward tactical insights through SHAP values, but the Platt-calibrated SVM yields the strongest overall classification performance (AUC: 0.6864). For an actual operational setup, the SVM model is the rational

choice because of its superior generalization, even though its non-linear RBF kernel is less intuitive to interpret than tree-based alternatives.

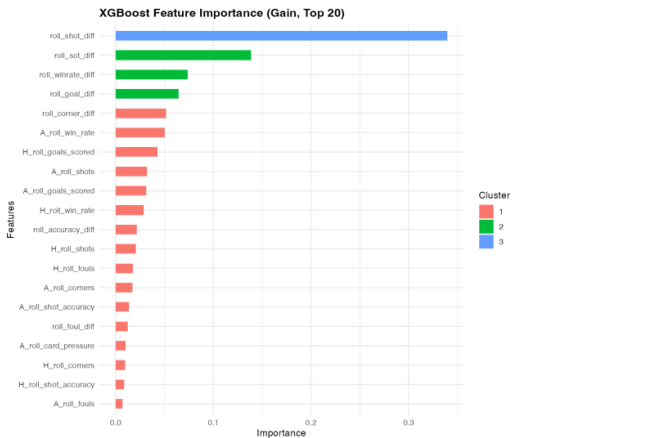


Fig. 10. SHAP summary plot for the XGBoost model, illustrating the global feature importance and the directional impact of the top predictive features.

B. Home Advantage Bias Analysis

Classification models always carry the risk of learning and amplifying structural biases present in the training data. Since home teams historically win more often in football, algorithms can easily develop a structural bias toward predicting the "Home Win" label. To quantify this, the False Positive to False Negative (FP/FN) ratio was utilized as a fairness metric.

On the test set, the model's False Positive rate (predicting a home win that did not happen) was 13.7%, while its False Negative rate (missing an actual home win) was 22.4%. This yields an FP/FN ratio of 0.613. Since this is well below 1.0, it clearly shows that the model does not suffer from a systematic home bias. In fact, it leans the other way, showing a conservative anti-home bias by under-predicting home victories. This implies that the model does not blindly overestimate the home advantage; instead, it puts significant weight on the away team's resilience.

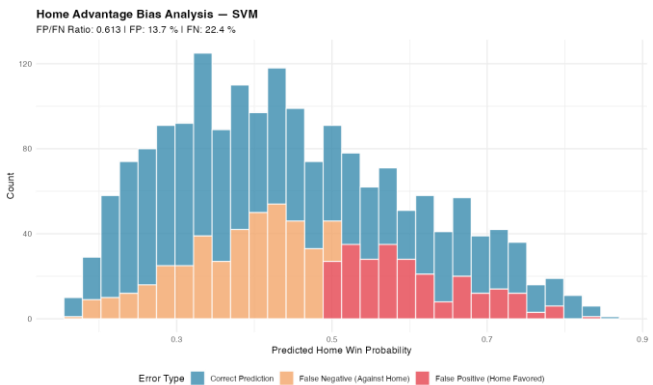


Fig. 11. Home advantage bias analysis displaying the distribution of error types across prediction probabilities. The FP/FN ratio of 0.613 indicates a conservative (anti-home) bias.

C. Reproducibility

To ensure complete reproducibility—a core ethical standard in scientific research—all random seeds were fixed at 42. All data manipulation and modeling steps were executed in the R programming language (version 4.3.1). Furthermore, the entire workflow was compiled into a modular pipeline utilizing the Expanding Window approach to strictly prevent any temporal data leakage.

VI. CONCLUSION

A. Summary of Findings

This study presented a time-aware machine learning pipeline designed to predict home victories in the English Premier League using strictly pre-match data. By replacing standard random k-fold splitting with a TimeSeriesSplit approach, the issue of temporal data leakage was resolved, making the generalization metrics much more reliable. Ultimately, a Platt-calibrated SVM model achieved the best overall results (AUC: 0.6864). However, the most interesting takeaway is how a simple probabilistic classifier like Naive Bayes managed to outperform heavy ensemble algorithms such as Random Forest and XGBoost. This practically demonstrates that in highly noisy and high-variance environments like football, simpler models can actually offer more robust generalization capabilities compared to complex ones.

B. Limitations

Naturally, the predictive power of this model is constrained by a few structural limitations. For one, it relies exclusively on historical rolling averages. Key contextual factors like real-time in-play events, sudden injuries, managerial sackings, and betting market movements were left out of the equation. Additionally, using a 5-match rolling window assumes that a team's true strength remains relatively constant during that specific period (the stationarity assumption). Lastly, external shocks like the COVID-19 pandemic, which forced games to be played behind closed doors, structurally altered the traditional home advantage and introduced temporary concept drift into the time series.

C. Future Work

Future studies could easily build upon the statistical foundation laid out here. Incorporating closing odds from financial betting markets as a Bayesian prior would likely provide a massive boost to the model's predictive signal. Also, instead of just looking at recent form, developing hierarchical team embeddings based on long-term squad quality and roster depth could capture underlying team strengths much better. Finally, exploring Ordinal Regression loss functions could be a great next step to more accurately model the inherent uncertainty surrounding the "Draw" class.

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